

$KG(8, 2)$ as a counterexample to coverage by two graphs of fractional chromatic number at most $5/2$

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Abstract

We prove that every cover of the edges of the Kneser graph $KG(8, 2)$ by two spanning subgraphs has a layer with fractional chromatic number at least $28/11$. Since $28/11 > 5/2$, this excludes a cover with both fractional chromatic numbers at most $5/2$. The proof reduces the question to the impossibility of a triangle-free edge partition whose two layers each contain an independent set of size twelve. Exact enumeration classifies the possible witnesses into 364 sets, and a Boolean branch certificate excludes all 728 cases remaining after a symmetry reduction. An independently implemented enumerator and certificate checker verify the finite argument. The bound is not claimed to be sharp.

1 Introduction

Gujgicz, Marits, and Ozeki study edge covers by graph families defined by bounds on graph parameters. In Section 4 and the first small-case question in Section 7 of their preprint [1], they ask whether $KG(8, 2)$ can be covered by two graphs of fractional chromatic number at most $5/2$. We answer this fixed question negatively. The same source distinguishes the fractional-coloring question from the existence of a cover by two triangle-free graphs. Our argument preserves this distinction: the obstruction requires large independent sets in *both* triangle-free layers.

We establish a quantitative lower bound through a finite combinatorial proof. The analytic part uses only the independent-set definition of fractional coloring. The computational part is an exhaustive classification followed by exact Boolean reasoning; it does not rely on a numerical linear-programming tolerance or on searching a finite list of fractional-coloring denominators. The cited source supplies context, and no external theorem is a premise of the proof below.

Independent sets of $KG(8, 2)$ are precisely intersecting families of two-subsets, so exact Erdős–Ko–Rado work provides adjacent structural context; a recent four-uniform result also supplies independently checked LRAT certificates for its exceptional small orders [2]. A complementary packing–cover certificate paradigm appears in exact work on Tuza’s conjecture for a restricted class of split graphs [3]. These are distinct theorems and neither is used in the proof below.

2 Definitions and endpoint

Write $[8] = \{1, \dots, 8\}$ and $K = KG(8, 2)$. Its vertex set is $V = \binom{[8]}{2}$, and two vertices are adjacent precisely when they are disjoint as subsets of $[8]$. Thus $|V| = 28$, each vertex has degree 15, and $|E(K)| = 210$. A host triangle consists of three pairwise disjoint pairs. There are $\binom{8}{6} \cdot 15 = 420$ such triangles.

For a finite graph G , let $\mathcal{I}(G)$ denote its independent vertex sets, including the empty set. Define

$$\chi_f(G) = \min \left\{ \sum_{I \in \mathcal{I}(G)} w_I : w_I \geq 0, \sum_{I \ni v} w_I \geq 1 \quad (v \in V(G)) \right\}.$$

Singleton sets give a feasible solution. The finite linear program attains its minimum: one may restrict to total weight at most $|V(G)|$, obtaining a nonempty compact feasible set. Empty sets need no positive weight. We permit arbitrary real weights, including zero.

A two-layer cover means spanning subgraphs H_0, H_1 of K with $E(H_0) \cup E(H_1) = E(K)$; overlaps are allowed. All isolated vertices are retained.

Theorem 1. *For all spanning subgraphs H_0, H_1 of $KG(8, 2)$,*

$$\begin{aligned} & \forall H_0, H_1 \text{ spanning subgraphs of } KG(8, 2), \\ & E(H_0) \cup E(H_1) = E(KG(8, 2)) \implies \\ & \max\{\chi_f(H_0), \chi_f(H_1)\} \geq \frac{28}{11} > \frac{5}{2}. \end{aligned}$$

Consequently no such cover has both fractional chromatic numbers at most $5/2$.

The quantifier in the display ranges over all pairs of spanning subgraphs. In words, every pair covering the host graph has at least one layer with fractional chromatic number at least $28/11$.

3 Reduction to independent sets

Lemma 2. *For every nonempty finite graph with N vertices and independence number α , $\chi_f(G) \geq N/\alpha$. A graph containing a triangle has fractional chromatic number at least 3. Deleting edges cannot increase fractional chromatic number.*

Proof. For any feasible weights, summing all vertex constraints yields

$$N \leq \sum_I |I| w_I \leq \alpha \sum_I w_I.$$

Here $\alpha \geq 1$, so division is legitimate. If T is a triangle, each independent set meets T in at most one vertex; summing only the three constraints for T gives $3 \leq \sum_I w_I$. Finally every independent set remains independent after edge deletion, so each old feasible coloring remains feasible. $\square$

Proposition 3. *There is no red–blue coloring of $E(K)$ with no monochromatic triangle in which each color layer has an independent set of size at least twelve.*

The next section gives the exact computational proof of this proposition.

Proof of Theorem 1 assuming Proposition 3. Choose, for each edge of an arbitrary cover, one containing layer. This produces a partition R, B by deleting edges from the layers. If either part contains a triangle, Lemma 2 gives a lower bound of $3 > 28/11$. Otherwise Proposition 3 implies that one part has independence number at most eleven. Applying the same lemma with $N = 28$ gives $\max\{\chi_f(R), \chi_f(B)\} \geq 28/11$. Edge-deletion monotonicity transfers this bound to the original cover. Finally, $28/11 - 5/2 = 1/22 > 0$, excluding the closed endpoint $5/2$ as well. $\square$

4 Exact finite certificate

4.1 Classification of witnesses

Identify a vertex of K with an edge of the complete graph on $[8]$. If a twelve-element set $A \subseteq V$ is independent in red and blue is triangle-free, A cannot contain three pairwise disjoint edges of the complete graph. Indeed, those three vertices would induce a blue triangle in K . An independent set larger than twelve contains a twelve-element subset, so this restriction loses no witnesses.

Let $\mathcal{A}$ be the twelve-element subsets of $\binom{[8]}{2}$ containing no matching of size three. Exact exhaustive enumeration gives

$$\mathcal{A} = \{D(P) \setminus \{e\} : P \in \binom{[8]}{2}, e \in D(P)\}, \quad D(P) = \{X \in \binom{[8]}{2} : X \cap P \neq \emptyset\}, \quad (1)$$

and $|\mathcal{A}| = 364$. Each $D(P)$ has thirteen elements. Every displayed set avoids a three-matching, since three disjoint edges would have to meet a set of only two vertices. Completeness of the displayed family is the enumerated part of the claim.

Here is the complete enumeration procedure. List the 28 pairs in lexicographic order. Recursively select increasing indices until twelve have been chosen, without pruning or symmetry reduction. At each leaf test containment of each of the 420 triples of pairwise disjoint pairs; reject the leaf if any is contained, and otherwise record it. The recursion visits each twelve-subset once, since its sorted index sequence is unique. It visits $\binom{28}{12} = 30,421,755$ leaves and records 364 distinct survivors. A separate checker constructs the right-hand side of (1), checks that it too contains 364 distinct sets, and compares the two sets of masks element by element with the serialized witness list. Thus classification is based on full coverage, not a sample or solely an agreement of counts.

4.2 Symmetry coverage

The deleted edge e in (1) either joins the two elements of P or joins one element of P to an element outside P . A permutation of $[8]$ therefore maps any first witness to one of

$$A_{\text{hub}} = D(\{1, 2\}) \setminus \{\{1, 2\}\}, \quad A_{\text{leaf}} = D(\{1, 2\}) \setminus \{\{1, 3\}\}.$$

Apply this permutation to the entire edge coloring and the second witness simultaneously. The second witness remains an arbitrary member of $\mathcal{A}$. Hence it suffices to examine two choices of the first witness and all 364 choices of the second: 728 cases. There is no separate normalization of the second witness, and coincident or overlapping witnesses are allowed.

4.3 Boolean clauses and branch verification

For every host edge e introduce a Boolean variable z_e , true for red. For every host triangle with edges e, f, g , impose

$$(z_e \vee z_f \vee z_g) \quad \text{and} \quad (\neg z_e \vee \neg z_f \vee \neg z_g).$$

These 840 clauses say exactly that both color layers are triangle-free. For a case (A, B) , force z_e false on every host edge internal to A , and true on every host edge internal to B . These initial assignments say exactly that A is red-independent and B is blue-independent. Conflicting initial assignments immediately exclude the case.

The certificate for each case is a binary tree. A leaf is encoded by `null`; an internal node is a triple consisting of a variable index and its false and true subtrees. The checker first applies unit propagation: a satisfied clause needs no action; an unsatisfied clause with no unset literal is a contradiction; an unsatisfied clause with exactly one unset literal forces that literal. It scans all

clauses repeatedly until no assignment changes. A conflict closes the current subproblem. Otherwise an internal node must name a legal, previously unset variable, and both assignments to that variable are recursively checked. A `null` leaf without a derived contradiction is rejected.

Soundness follows by induction on the tree. Unit propagation preserves every possible satisfying extension. A conflict admits no satisfying extension. At a branch every satisfying extension would belong to one of the two child assignments, both excluded by induction. If propagation already gives a conflict at an internal node, its unused descendants need not be checked. The checker also verifies that the root keys are exactly the 728 required pairs, with no repetition or omission.

The independently implemented replay closes all 728 roots. It visits 66,358 nodes, comprising 32,815 branch nodes and 33,543 conflict endpoints. The identity $66,358 = 728 + 2 \cdot 32,815$ checks the binary-forest counts. These counts describe the checked certificates; they are not substituted for clause evaluation. Every possible pair of witnesses has now been excluded, proving Proposition 3 and completing the theorem.

5 Reproducibility and computational assistance

The accompanying `supplement/` directory contains the exact witness list, branch trees, an independent C++ enumerator, and an independent Python checker. The programs label the ground set $0, \dots, 7$; adding one gives the notation used here. Vertex indices are the lexicographically ordered pairs, edge indices are lexicographically ordered pairs of disjoint vertices, and a subset is stored as the sum of 2^i over its vertex indices. Certificate rows store the representative type, the second witness mask, and the tree.

From `supplement/`, the following commands regenerate the exhaustive list and replay the verification. A C++17 compiler and Python 3 with its standard library suffice; Python assertions must remain enabled.

```
c++ -O3 -std=c++17 enumerate.cpp -o enumerate
./enumerate > masks.txt 2> enumeration.json
python3 check.py
```

The enumeration output must report 30,421,755 subsets, 364 survivors, and 420 forbidden triples. The checker compares classification sets, replays all roots, and writes its numerical summary to `result.json`. Verification uses only integer arithmetic, finite sets, and Boolean logic for the decisive argument. There are no floating-point tolerances.

Discovery used SAT encodings, solver searches, and weighted fractional-coloring models. Some stronger or alternative searches timed out; a timeout supplies no mathematical conclusion. Solver-generated unsatisfiability reports and DRAT traces were retained during discovery, but the proof presented here uses the separately checked branch trees instead. The independent mathematical audit did not certify those unused DRAT traces. The independent enumerator uses increasing-index recursion, and the independent checker uses integer bit masks and full clause scans; neither imports the producer’s decisive implementation.

Two negative tests reject unjustified conflict leaves, including truncation of an actual root that needs branching. Additional exact controls verify a triangle-free coloring with a twelve-element independent set in one layer, and a rational fractional-coloring cover for $KG(7, 2)$. The latter uses exact rational arithmetic. These controls test the implementation and the scope of the finite proposition; they are not premises needed to prove the endpoint. In particular, the one-layer control prevents interpreting the proposition as forbidding a large independent set in just one layer.

6 Limitations and disclosure

The theorem concerns this fixed host graph and two spanning layers. It gives no classification of extremal covers and does not prove that $28/11$ is attainable or optimal. It does not resolve the broader asymptotic questions concerning Kneser graph covers. No conclusion is drawn from the unsuccessful search for two independent eleven-element witnesses or from a timed-out weighted model.

This is a computer-assisted proof: exhaustive classification and branch replay form part of its evidence. The implementation, compiler, interpreter, and underlying hardware are within its computational trust boundary; no proof-assistant formalization is claimed. The independent mathematical check reconstructs the finite reduction and verification, but it is not a journal refereeing decision or an audit of the entire literature. The contextual attribution above is to the specified preprint version; no exhaustive priority or global novelty claim is made.

Language-model assistance was used in mathematical exploration, development of computational models and verification code, independent-context checking, and preparation of the manuscript. The independent check refers to a separate computational and language-model context, not to an identified external human referee. Model-generated assertions and solver status messages are not treated as proof: the result rests on the stated mathematical deductions and exact finite checks. The author identity is supplied with the project; no additional human verification or endorsement is asserted here.

References

- [1] Anna Gujgicz, Márton Marits, and Kenta Ozeki. Cover numbers by graph families bounded by certain graph parameters, 2026. Version 1, preprint.
- [2] Zijian Zeng. A sharp 3-degree Erdős–Ko–Rado theorem for 4-uniform families. SSRN preprint, 2026.
- [3] Zijian Zeng. Tuza’s conjecture for split graphs with an eight-vertex clique part and two neighborhood types. Preprints.org, August 2026. Preprint, version 1, posted 19 August 2026.